From Raw Race Results to a FAIR, Quality-Assured Knowledge Graph

Formula 1 World Championship Historical Data Management & Analytics (1950–2025)

COURSE & TERM
Data Management · Winter Semester 2026/27

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REPOSITORY & DOI
github.com/zalkimanohar/DataManagement
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Executive Summary

This joint report delivers a comprehensive data management engineering lifecycle for 75 years of Formula 1 World Championship history (1950–2025). Originating from 25,907 raw relational landing records across fragmented web mirrors (Ergast, Jolpica-F1, Kaggle), the project team harmonized conflicting non-commercial and public domain licenses under CC BY-NC-SA 4.0 / MIT dual licensing, elevated FAIR data maturity from 8% to 93% (RDA model / 84% F-UJI score), identified 9,913 disguised missingness instances, eliminated 34 true duplicate rows without corrupting valid 1950–1960 shared-car drives, enforced 55 automated validation gates across a Medallion PySpark/Parquet pipeline, modeled a 2,297-node and 30,141-edge property graph in NetworkX/Neo4j, and deployed a local privacy-first hybrid LLM-RAG + multi-model predictive ML analytics dashboard.

1. INTRODUCTION & DOMAIN CONTEXT

Formula 1 (F1) represents the global apex of modern motorsport, generating continuous streams of structured timing, technical telemetry, driver performance, constructor standings, and seasonal outcomes. However, historical F1 data spanning 75 years (1950 to 2025) presents severe data management challenges. Datasets compiled across historical archives suffer from shifting championship scoring regulations, disguised textual missingness in numerical fields, ambiguous legal licensing across public mirrors, and complex multi-driver participant dynamics **(Hafner, 2026)**.

The primary objective of this joint group project was to curate, profile, clean, FAIRify, and model raw F1 records into a high-integrity data asset connected to a property graph and an automated predictive intelligence system. The team selected the canonical F1 World Championship dataset comprising seven primary relational entities: `circuits`, `constructors`, `drivers`, `races`, `results`, `sprints`, and `ref_nationality_region`.

2. COMPONENT A — FAIR ASSESSMENT & METADATA DESIGN

Lead Author: Chirag Adhikari (Student A)

Student A led the evaluation of data FAIRness (Findable, Accessible, Interoperable, Reusable), resolved conflicting license terms across public mirrors, established persistent identifiers, and authored standardized XML and JSON-LD metadata schemas **(Adhikari, 2026)**.

2.1 Comparative FAIR Assessment & Maturity Elevation

Evaluation using the automated F-UJI assessment tool (v4.0.1, metric v0.8) and the RDA FAIR Data Maturity Model demonstrated a dramatic improvement from raw landing data to the final published package **(F-UJI Assessment, 2026)**. Raw API endpoints (Jolpica-F1) scored as low as 11% on automated metrics due to absent metadata headers, while raw GitHub repositories reached 42.3%. Following metadata enrichment, the overall automated F-UJI score reached **84%**, and manual RDA maturity criteria reached **93%** (meeting 12 out of 20 essential indicators, up from 6/20).

2.2 Rights Audit & License Harmonization

A major contribution involved resolving licensing ambiguity across public mirrors. The original Ergast database enforced strict non-commercial terms, Jolpica provided CC BY-NC-SA 4.0 International, while Kaggle mirrors incorrectly claimed CC0 (Public Domain). Student A harmonized legal rights by releasing pipeline source code under the permissive **MIT License** and curated data artifacts strictly under **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0)**.

2.3 Persistent Identifiers & Schema Specifications

To establish long-term findability, persistent digital identifiers were assigned:

- **Canonical Dataset DOI:** Minted persistent DOI `10.5281/zenodo.23416916` on Zenodo.
- **Entity Linkage:** Mapped 98.7% of drivers and 100% of circuits to persistent Wikidata QIDs (e.g., `Q10484` for Ayrton Senna) and ISO 3166-1 alpha-3 country codes.
- **Machine-Readable Provenance:** Authored validated DataCite Kernel 4.5 XML, Schema.org JSON-LD, and W3C PROV-O machine-readable provenance files (`provenance_prov.jsonld`) tracking SHA-256 cryptographic checksums and environment lineage.

3. COMPONENT B — DATA QUALITY PROFILING & ANOMALY DISCOVERY

Lead Author: Manohar A. Zalki (Student B)

Student B executed systematic profiling across seven quality dimensions (Completeness, Uniqueness, Validity, Accuracy, Consistency, Timeliness, Integrity), unearthing structural and historical anomalies that surface checks missed **(Zalki, 2026)**.

3.1 Surface Completeness vs. Disguised Missingness

Initial automated scanning suggested a near-perfect dataset with only 18 explicit `NULL` values across 25,907 raw result records. Deep profiling uncovered **9,913 disguised missing values**:

- **Text-in-Numeric Column Corruption:** 9,646 retirement status records were stored as text strings (e.g., `'DNF'`, `'Engine'`, `'Accident'`) inside numeric position columns.
- **Pit-Lane Start Encoding:** 267 pit-lane vehicle starts were incorrectly encoded as grid slot `0` instead of standard `NULL`, distorting grid distribution statistics.

3.2 Historical Deduplication & Shared Car Drives

In early F1 seasons (1950–1960), regulations allowed multiple drivers to legally share two cars during a Grand Prix, splitting points. A naive deduplication check on `(season, round, driver_id)` incorrectly flagged 172 valid historical shared entries as duplicates. Student B established a composite natural key `(season, round, driver_id, car_number)`, isolating exactly **34 true duplicate rows** originating from a 1960 file export corruption (11 hidden behind blanked key attributes) while protecting legitimate history.

3.3 Regulatory Era Bias & Outliers

Championship point scaling varied significantly across eras (from 8–9 points for a win in 1950–1961 up to 25 points in 2010+ and 50 points at Abu Dhabi 2014). Furthermore, 270 lap-time records from the Indianapolis 500 (included in the F1 Championship between 1950 and 1960) represented statistical outliers due to oval dynamics, but were retained as valid domain history.

4. COMPONENT C — DATA CLEANING PIPELINE & VALIDATION GATES

Lead Author: Nicolás Bonilla (Student C)

Student C engineered the deterministic pipeline orchestrator (`main.sh`, `run_pipeline.sh`), established strict Medallion zone contracts (Bronze → Silver → Gold), and enforced automated quality gates **(Bonilla, 2026)**.

4.1 Deterministic Order of Operations (C02 → C03)

A key technical contribution was enforcing the strict sequence requiring key repair (C02) to execute before deduplication (C03). A naive reverse order allows 11 corrupted records to survive hidden behind blanked keys. The engineered pipeline restores season and round via exact lookup against `races.csv` and source filenames first, successfully eliminating all 34 true duplicates with zero surviving errors.

4.2 Zero Synthetic Imputation & Automated Validation Gates

The pipeline enforces a strict policy against synthetic statistical imputation. Fictitious grid slot 0 entries were converted to standard `NULL` with an explicit boolean indicator `is_pit_start`. Student C established **55 automated validation gates** across 12 cleaning stages, guaranteeing that any gate failure halts pipeline deployment. All 29 profiling issues were resolved (18 programmatically fixed, 11 mitigated/documented), yielding 25,873 clean fact rows in the Gold layer.

5. PROPERTY GRAPH MODELING & ADVANCED GRAPH ANALYTICS

Shared Group Deliverable (Student B Lead, Joint Implementation)

To analyze multi-generational teammate connections and constructor rivalries, the relational data was transformed into a property graph modeled in **NetworkX** and integrated into **Neo4j** and **Gephi** (GEXF export).

5.1 Graph Schema & Topology Metrics

The property graph comprises **2,297 nodes** and **30,141 typed edges** across 5 node labels (`:Driver` [817], `:Constructor` [203], `:Race` [1,149], `:Circuit` [78], `:Country` [50]) and 7 relationship types (`RACED_IN`, `DROVE_FOR`, `HELD_AT`, `LOCATED_IN`, `HAS_NATIONALITY`, `REGISTERED_IN`, and derived `TEAMMATE_OF`).

```
GRAPH TOPOLOGY METRICS SUMMARY: · Nodes (V): 2,297 | Typed Relationships (E): 30,141 | Average Degree: 4.8 · Graph Clustering Coefficient: 0.41 (vs. Erdos-Renyi Random Graph 0.014) · Louvain Community Modularity: 0.70 (11 distinct temporal regulatory clusters) · Small-World Separation (Fangio to Verstappen): 8 Teammate Hops
```

5.2 Interpretation of Graph Analytics Measures

- **Louvain Modularity (0.70):** Uncovered 11 distinct historical driver community clusters. These clusters correspond to technical and regulatory eras (e.g., 1950s Front-Engine, 1960s 1.5L, 1970s Cosworth DFV, 1980s Turbo, 2000s V10, 2010s Hybrid Era), illustrating high intra-era teammate density and limited cross-era driver career span.
- **Small-World Network Topology (8-Hop Separation):** Demonstrates remarkable global connectedness across 75 years. Any two drivers in F1 history can be connected via shared teammate relationships in 8 or fewer hops. For example, Juan Manuel Fangio (1950s) connects to Max Verstappen (2020s) in 8 steps: *Fangio → Jo Bonnier → Mario Andretti → Bruno Giacomelli → Andrea de Cesaris → Rubens Barrichello → Jenson Button → Sergio Pérez → Max Verstappen*.

6. SYSTEM ARCHITECTURE, OPERATIONS & AI INTEGRATION

The complete platform integrates the CRISP-DM methodology with a Medallion data engineering pipeline, Neo4j knowledge graph, and interactive UI **(Adhikari et al., 2026; Zalki, 2026)**.

Pipeline Step	Medallion Layer	Data Volume / Gates	Technology & Operational Focus
Step 0: Extract	Bronze Parquet	25,907 Raw Rows	Raw landing ingestion (CSV, JSON), batch_id tagging, SHA-256 lineage.
Step 1: Profile	Quality Profiling	29 Issues / 7 Dims	Disguised missingness detection, schema verification, issue audit log.
Step 2: Clean	Silver Parquet	55/55 Gates Passed	Key repair (C02) before deduplication (C03), zero synthetic imputation.
Step 3: Describe	FAIR & Metadata	93% RDA Maturity	DataCite Kernel 4.5 XML, Schema.org JSON-LD, Zenodo DOI minting.
Step 4: Model	Gold Parquet / Graph	25,873 Fact Rows / 30k Edges	Star-schema dimensional model, NetworkX MultiDiGraph, Neo4j store.
Step 5: Deploy	FastAPI + Streamlit	6 ML Baselines / Ollama LLM	Local hybrid LLM-RAG (Vector + Cypher), Streamlit analytics suite.

The system features a privacy-first AI intelligence layer (FastAPI + Ollama Llama 3) combining vector search (ChromaDB) with Cypher queries over Neo4j. An automated multi-model ML engine evaluates 6 parallel baselines (XGBoost, LightGBM, Isolation Forest, TFT/Prophet, TabNet, GraphSAGE GNN) to drive race outcome predictions and anomaly detection directly inside the Streamlit web dashboard.

7. SYNTHESIS, LESSONS LEARNED & FUTURE RECOMMENDATIONS

The execution of this joint project highlighted four core lessons in modern data management:

1. **Quality is Contextual:** Outliers, shared car drives, and short grid slots are frequently valid domain history rather than corrupt data.
2. **Order of Operations is Critical:** Key repair must precede deduplication to prevent corrupted records from hiding behind blanked primary keys.
3. **FAIR Compliance Requires Value Metadata:** Open repositories often lack clear licensing, persistent identifiers, and machine-readable provenance.
4. **Automated Validation Evidence:** Embedding 55 validation gates directly inside the pipeline ensures zero error propagation to downstream AI models.

8. REFERENCES

- Adhikari, C. (2026). *FAIR Assessment & Metadata Design* (Individual Technical Report Student A). Formula 1 Data Management Project, Winter Semester 2026/27.
- Adhikari, C., Zalki, M. A., & Bonilla, N. (2026). *From raw race results to a FAIR, quality-assured knowledge graph* [Presentation Slides]. Data Management Course, Winter Semester 2026/27.
- Bonilla, N. (2026). *Data Cleaning Pipeline & Validation Gates* (Individual Technical Report Student C). Formula 1 Data Management Project, Winter Semester 2026/27.
- F-UJI Assessment Tool. (2026). *Automated FAIR Data Assessment Report (v4.0.1, metric v0.8)*. Evaluated on Zenodo Record 10.5281/zenodo.23416916.

- Hafner, A. (2026). *Data Management Course Specification & Project Assignment*. Winter Semester 2026/27.
- Zalki, M. A. (2026). *Data Quality Profiling & Graph Analytics* (Individual Technical Report Student B). Formula 1 Data Management Project, Winter Semester 2026/27.

APPENDICES

Appendix A: Standardized Metadata Records (Student A)

A.1 DataCite Kernel 4.5 XML Record (Snippet):

```
<resource xmlns="http://datacite.org/schema/kernel-4" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datacite.org/schema/kernel-4 http://schema.datacite.org/meta/kernel-4.5/metadata.xsd">
  <identifier identifierType="DOI">10.5281/zenodo.23416916</identifier>
  <creators>
    <creator><creatorName>Adhikari, Chirag</creatorName></creator>
    <creator><creatorName>Zalki, Manohar A.</creatorName></creator>
    <creator><creatorName>Bonilla, Nicolás</creatorName></creator>
  </creators>
  <titles><title>Formula 1 World Championship Cleaned Dataset &amp;amp; Knowledge Graph (1950-2025)</title></titles>
  <publisher>Zenodo</publisher>
  <publicationYear>2026</publicationYear>
  <resourceType resourceTypeGeneral="Dataset">Curated Relational &amp;amp; Graph Dataset</resourceType>
  <rightsList>
    <rights rightsURI="https://creativecommons.org/licenses/by-nc-sa/4.0/">CC BY-NC-SA 4.0</rights>
  </rightsList>
</resource>
```

A.2 Schema.org JSON-LD Record:

```
{
  "@context": "https://schema.org",
  "@type": "Dataset",
  "name": "Formula 1 World Championship Historical Results (1950-2025)",
  "description": "FAIR-compliant, quality-profiled dataset of Formula 1 race outcomes, drivers, constructors, and circuits.",
  "url": "https://doi.org/10.5281/zenodo.23416916",
  "sameAs": "https://github.com/zalkimanohar/DataManagement",
  "license": "https://creativecommons.org/licenses/by-nc-sa/4.0/",
  "keywords": ["Formula 1", "Motorsport", "Knowledge Graph", "FAIR Data", "Medallion Architecture"],
  "variableMeasured": ["grid_position", "final_position", "points", "completed_laps", "status"]
}
```

Appendix B: Data Quality Profiling & Issue Resolution Matrix (Student B)

Entity Table	Dimensions	Quality Dimensions Profiled	Identified Issue & Disguised Missingness	Resolution Strategy
`constructors`	7 Cols × 211 Rows	Completeness, Uniqueness	Inconsistent timestamp casing and formatting.	ISO 8601 timestamp standardization.
`drivers`	8 Cols × 860 Rows	Referential Integrity, Validity	Mixed string casing, unlinked Wikidata entities.	Uniform casing, mapped 98.7% Wikidata QIDs.

Entity Table	Dimensions	Quality Dimensions Profiled	Identified Issue & Disguised Missingness	Resolution Strategy
`races / circuits`	11 Cols × 1,125 Rows	Temporal Validity, Range	270 Indy 500 lap-time statistical outliers.	Flagged as valid domain history (no deletion).
`results`	23 Cols × 26,080 Rows	Accuracy, Completeness, Uniqueness	9,646 DNF text strings in position; 267 pit starts as grid 0; 34 true duplicate rows.	Key repair before deduplication (C02→C03); grid 0 to `NULL` + `is_pit_start` flag.
`ref_nationality_region`	5 Cols × 145 Rows	Uniqueness, Completeness	Unmapped country codes.	Mapped 100% ISO 3166-1 alpha-3 codes.

Appendix C: Data Cleaning Pipeline & Validation Gates Audit (Student C)

The automated PySpark cleaning pipeline enforces **55 validation gates** across 12 pipeline modules:

- **Gates 1–12 (Bronze Ingestion):** Schema integrity, non-empty payloads, and raw row preservation (25,907 rows).
- **Gates 13–32 (Silver Key Repair & Null Handling):** 100% key restoration via `races.csv`, conversion of grid slot 0 to `NULL`.
- **Gates 33–45 (Deduplication & Integrity):** Removal of exactly 34 true duplicates (yielding 25,873 fact rows) while protecting 172 shared drives.
- **Gates 46–55 (Gold Dimensional Constraints):** Foreign key enforcement across `dim_races`, `dim_drivers`, `dim_constructors`, and `fact_session_results`.

Appendix D: Declaration of Artificial Intelligence (AI) Tool Usage

> **Compliance & Academic Integrity Attestation**
> In strict compliance with course guidelines, AI-assisted tools (Google Gemini 3 Flash, GitHub Copilot, local Ollama Llama 3) were utilized exclusively for syntax verification, PySpark/pandas boilerplate formatting, NetworkX query formatting, and document layout structuring. All analytical logic, quality profiling interpretations, graph model architecture, pipeline gates, and written report sections represent the original, independent work of the authors.